

## 6 Normal distribution

| Subject content                                                                                                                                                                                            | Additional information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 6.1 Know the specific properties of the normal distribution, and know that data from such an underlying population would approximate to having these properties, with different samples showing variation. | Students should be aware that the normal distribution has a bell-shaped probability density curve.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6.2 Apply knowledge that approximately $\frac{2}{3}$ of observations lie within $\mu \pm \sigma$ , and equivalent results for $2\sigma$ and $3\sigma$ .                                                    | Students should learn that approximately 95% of observations lie within $\mu \pm 2\sigma$ and approximately 99.8% of observations lie within $\mu \pm 3\sigma$                                                                                                                                                                                                                                                                                                                                         |
| 6.3 Determine probabilities and unknown parameters with a normal distribution.                                                                                                                             | Students may use calculator functions to obtain information for a normal distribution directly and are advised to do so.                                                                                                                                                                                                                                                                                                                                                                               |
| 6.4 Apply the normal distribution to model real-world situations.                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6.5 Use the fact that the distribution of $\bar{X}$ has a normal distribution if $X$ has a normal distribution.                                                                                            | Students are expected to know that $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6.6 Use the fact that the normal distribution can be used to approximate a binomial distribution under particular circumstances.                                                                           | Students should learn that, as an alternative to using the exact binomial distribution, the normal distribution may be used to approximate a binomial distribution when: <ul style="list-style-type: none"> <li>• <math>n \geq 20</math> and <math>p \approx 0.5</math></li> <li>• <math>np &gt; 10</math> or <math>n(1-p) &gt; 10</math></li> </ul> Students will be expected to use a continuity correction when finding a probability if a normal approximation to a binomial distribution is used. |